GENERAL APTITUDE

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Calendar

Q. What was the day of the week on 15th August, 1947?

Soln:

Completed till 1946

1946

$\frac{1900}{400} = 300$ → 1 odd day

$\frac{46}{4} = 11$(quotient)

46 + 11 = 57 $\frac{57}{7} = 1$(remainder)

In 1946, odd days are,

1900 46

1 + 1 = 2 odd days

	1946		month		date		
Total odd days =	2	+	2	+	1	=	5 odd days

As per table for days of a week , 5 ⟷ Friday

As month is August, go till July as per table,

J		F		M		A		M		J		J	
3	+	0	+	3	+	2	+	3	+	2	+	3	= 16

Now, $\frac{16}{7} = 2$ (remainder)

For date ,

$\frac{15}{7} = 1$ (remainder)

Calendar

For Months -

J	F	M	A	M	J	J	A	S	O	N	D
0	3	3	6	1	4	6	2	5	0	3	5

For years -

1600 – 1699	6
1700 – 1799	4
1800 – 1899	2
1900 – 1999	0
2000 – 2099	6

Calendar

Q. What day of the week was 29 June 2010 ?

- **Soln**

1. Last 2 digits of the year → 10
2. Divide by 4 (10 ÷ 4) = 2(quotient)
3. Take the date → 29
4. Take the no. of month → 4(from table)
5. Take the no. of year → 6 (from table)

51 (add)

6. Divide by 7 → $\frac{51}{7}$ = 2(remainder)

Check table for day of the week

2 ⟷ Tuesday

Calendar

Q. What was the day of the week on 29th February, 2012?

Soln:

1. Last 2 digits of the year → 12
2. Divide by 4 (12 ÷ 4) = 03(quotient)
3. Take the date → 29
4. Take the no. of month → 03 (from table)
5. Take the no. of year → 06 (from table)

53 (add)

6. Divide by 7 → $\frac{53}{7}$ = 4 (remainder)

subtract 1 from remainder

In this case for all dates of **January & February** in a leap year , 4 -1 =3

Check table for day of the week

3 ⟷ Wednesday

Calendar

It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?

A. Sunday B. Saturday C. Friday D. Wednesday

Ans: C

On 31st December, 2005 it was Saturday.

Number of odd days from the year 2006 to the year 2009 = (1 + 1 + 2 + 1) = 5 days.

On 31st December 2009, it was Thursday.

on 1st Jan, 2010 it is Friday.

Calendar

Q. If we have preserved the calendar of 2017. Find the next immediate year in which we can reuse.

A. 2027 B.2023 C. 2025 D. 2029

Soln:

x/4 (x = given year)

$\frac{2017}{4}$ = 1 (remainder)

For any year divide by 4, the possibility of remainder is 0,1,2,3

If remainder = 0 → x + 28

If remainder = 1 → x + 6

If remainder = 2/3 → x + 11

So, $\frac{2017}{4}$ = 1(remainder)

2017 + 6 = 2023 **Ans: B**

Calendar

Q. Which of the following days can never be the last day of a century?

A. Sunday B. Monday C. Tuesday D. Wednesday

- **Soln**:
- The last day of century can be only
- 1 odd day(Monday)
- 3 odd days (Wednesday)
- 5 odd days (Friday)
- 7 or 0 odd days (Sunday)
- So, century can never end in **Tuesday** , **Thursday** or **Saturday**.
- **Ans: C**

Calendar(Assignment)

Q. What was the day of the week on 26^{th} January, 1947?

Soln:

1. Last 2 digits of the year → 47
2. Divide by 4 (47 ÷ 4) = 11(quotient)
3. Take the date → 26
4. Take the no. of month → 0 (from table)
5. Take the no. of year → 0 (from table)

84 (add)

6. Divide by 7 → $\frac{84}{7}$ = 0 (remainder)

Check table for day of the week

0 ⟷ Sunday

Calendar(Assignment)

- Q. The day on 5th April of a year will be the same day on 5th of which month of the same year?
- A. 5th July B. 5th August C. 5th June D. 5th October

- **Ans A**

- April & July for all years have the same calendar. So, a day on any date of April will be the same day on the corresponding date in July.
- The same day will fall on 5th July of the same year.

Calendar(Assignment)

Q. What was the day of the week on your birthdate?

Q. 13th October 2019 is a Sunday. Find the day on 13th October 1989?

A. Sunday B. Monday C. Friday D. Wednesday

Ans: C

Q. 1st March 2006 falls on a Wednesday .What day does 1st March 2010 fall on?

A. Tuesday B. Monday C. Friday D. Wednesday

Ans: B

Q. Today is Monday. Which day will be after 64 days?

A. Tuesday B. Monday C. Friday D. Wednesday

Ans: A

Q. Today is Monday. After 30 days it will be?

A. Tuesday B. Monday C. Friday D. Wednesday

B. Ans: D

Calendar(Assignment)

Q. 15th August 1947 was a Friday. Find the day on 15th August 1977?

- Soln:

$$\begin{array}{r} 1977 \\ -\ 1947 \\ \hline 30 \text{ years} \end{array}$$

Leap years between 1947 to 1977

1948	1964	
1952	1968	8 years
1956	1972	
1960	1976	

30 + 8 = 38

total years leap

$\frac{38}{7}$ = 3 (remainder)

As 15th August 1947 was a Friday ,

So, Friday + 3 days = **Monday**

Calendar(Assignment)

Q. 4th January 2016 falls on Monday. What day of the week does 4th January 2017 lies?

A. Wednesday B. Thursday C. Tuesday D. Monday

Soln:

Normal year = 1 odd day

Leap year = 2 odd days

Jan 4, 2016 → Monday

\+ 2 (as leap year)

Jan 4,2017 → Wednesday

Ans: A

Calendar(Assignment)

Q. Wednesday falls on 5th of a month .So which day will fall 5 days after 22^{nd} of the same month?

A. Tuesday B. Friday C. Thursday D. Wednesday

Ans: B

5^{th} = Wednesday

+7

12^{th} = Wednesday

+7

19^{th} = Wednesday

22^{nd} = Saturday

+5

27^{th} = Thursday

5 days after 22^{nd} will be **Friday**

Calendar(Assignment)

Q. What dates of May 2002 did Monday fall on?

Soln:

Lets take date = 1st May 2002

1. Last 2 digits of the year → 02
2. Divide by 4 (02 ÷ 4) = 00(quotient)
3. Take the date → 01
4. Take the no. of month → 01 (from table)
5. Take the no. of year → 06 (from table)

10 (add)

6. Divide by 7 → $\frac{10}{7}$ = 3 (remainder)

Check table for day of the week

3 ⟷ Wednesday

1st May 2002 falls on Wednesday

1	2	3	4	5	6
W	Th	F	Sa	Su	M

first Monday

Now add 7 to it to find remaining Mondays

Dates on which Monday falls are -

6 , 13 , 20, 27

Calendar(Assignment)

Q. On what dates of April, 2001 did Wednesday fall?

A. 1st, 8th, 15th, 22nd, 29th B. 2nd, 9th, 16th, 23rd, 30th C. 3rd, 10th, 17th, 24th

D. 4th, 11th, 18th, 25th

Ans: D

Calendar(Assignment)

Q. What is the day on 22 April 2222?

A. Monday B. Tuesday C. Saturday D. Sunday

Ans: A

Calendar(Assignment)

Which of the following is not a leap year?

A. 700 B. 800 C. 1200 D. 2000

Ans: A

The century divisible by 400 is a leap year.
The year 700 is not a leap year.

Calendar(Assignment)

Q. Today is Monday. Which day will be on 61st day?

Soln:

1 week = 7 days. Taking the multiple of 7

56 - Monday	or	63 - Monday
57 – Tuesday		62 - Sunday
58 – Wednesday		61 - Saturday
59 – Thursday		
60 – Friday		
61 - Saturday		
56 + 5 = 61 days		63 – 61 = 2 days
(add 5 days)	or	(subtract 2 days)

Calendar(Assignment)

Q. January 1, 2007 was Monday. What day of the week lies on Jan. 1, 2008?

A. Monday

B. Tuesday

C. Wednesday

D. Sunday

Ans: B

THANK YOU!

Mixtures & Alligation

- **Alligation** : It is the rule which enables us to find the ratio in which two or more ingredients at given prices must be mixed to produce a mixture of a desired price.(mixing / linking)
- **Mean Price** : The cost price of a unit quantity of mixture is called the mean price.
- **Dearer** : The more expensive ingredient

- Note :

Always maintain the order in which problem is given else answer gets changed

Mixtures & Alligation

Type 1 oranges at Rs.60 per kg and Type 2 oranges at Rs.120 per kg and when mixed cost is Rs.75 per kg. Find the ratio in which Type 1 and Type 2 oranges are mixed.

Soln:

Type 1		Type 2
60		120
	75	
x = d-m		y =m-c

$$\frac{x}{y} = \frac{d-m}{m-c} = \frac{120-7}{75-6} = \frac{45}{15} = \frac{3}{1} = 3:1$$

CP of cheaper ingredient (c)

CP of costlier ingredient (d)

Mean Price (m)

CP of costlier ingredient - Mean Price

Mean Price - CP of cheaper ingredient

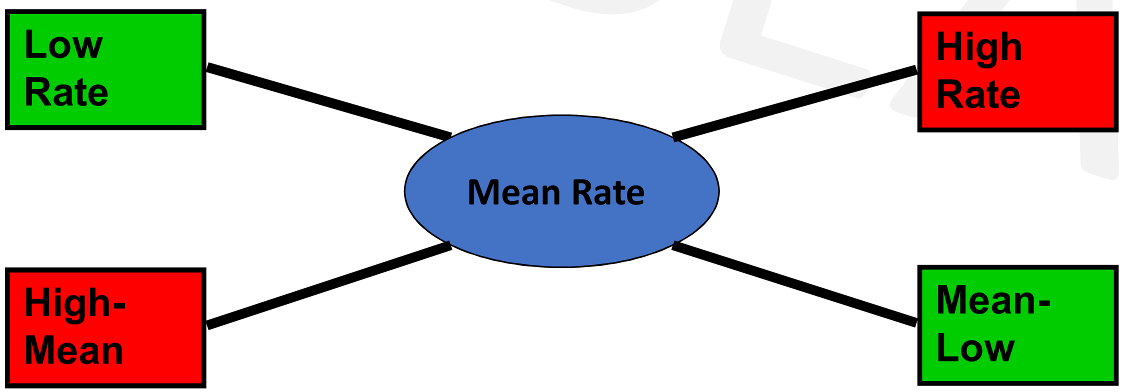
$$\frac{\text{Quantity of cheaper ingredient}}{\text{Quantity of costlier ingredient}} = \frac{d - m}{m - c}$$

Mixtures & Alligation

$$\frac{\text{Quantity of Lower}}{\text{Quantity of Higher}} = \frac{(\text{C.P. of Higher}) - (\text{Mean Price})}{(\text{Mean Price}) - (\text{C.P. of Lower})}$$

$$\frac{Ql}{Qh} = \frac{CPh - CPm}{CPm - CPl}$$

(**Qty** Low) : (**Qty** High) = (**CP**h-**CP**m) : (**CP**m-**CP**l)



Mixtures & Alligation

Q. In what ratio must a grocer mix two varieties of pulses costing Rs. 15 and Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.50 per kg ?

A. 3 : 7

B. 5 : 7

C. 7 : 3

D. 7 : 5

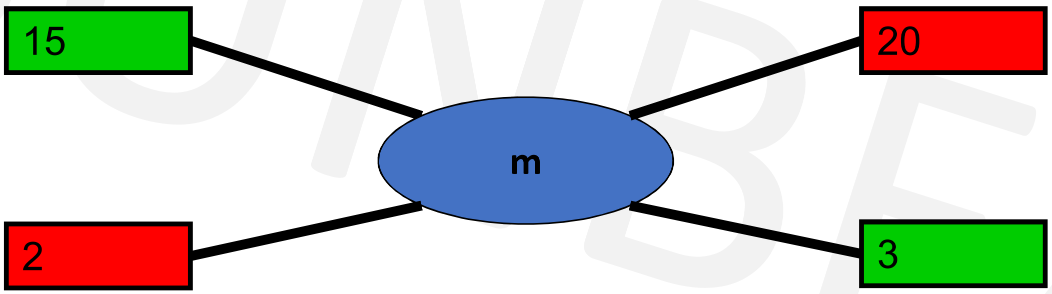
Ans: C

Mixtures & Alligation

Q. CP of rice A is Rs. 15/kg and CP of rice B is Rs.20/kg. If both A and B are mixed in the ratio 2:3. Then find the price per kg of the mixed rice.

A. Rs. 28 B. Rs. 17 C. Rs. 18 D. Rs. 48

Soln:



$$\frac{x}{y} = \frac{d-m}{m-c}$$

$$\frac{2}{3} = \frac{20-m}{m-15}$$

$$m = \frac{90}{5} = Rs.18$$

Ans: C

Mixtures & Alligation

Q. In what ratio must a grocer mix two varieties of dal worth Rs. 60/kg & Rs. 65/kg, so that selling the mixture at 68.20/kg, he may gain 10%.

Soln:

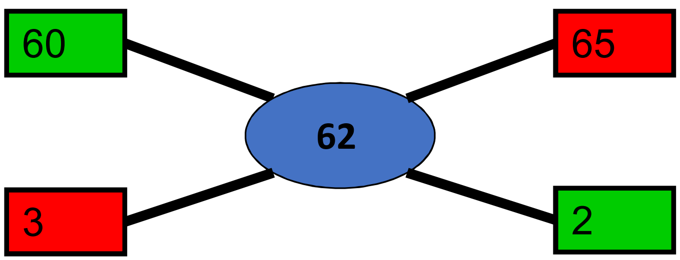
- Mean price is always CP
- Steps-
- 1. m=?
- 2. m = cost price(CP)
- 3. SP = given
- 4. find x/y=?

Mixtures & Alligation

In what ratio must a grocer mix two varieties of dal worth Rs. 60/kg & Rs. 65/kg, so that selling the mixture at 68.20/kg, he may gain 10%.

A. 3:2　　B. 2:3　　C. 3:4　　D. 4:3

- SP of 1 kg of mixture = Rs. 68.20
- Gain =10%
- In case of profit, SP = $\frac{\text{C.P. x } \mathbf{(100 +\%gain)}}{100}$
- CP of 1kg of mixture = Rs ($\frac{100}{100+10}$ x 68.2)
- $= \frac{682}{11}$
- Mean price =Rs. 62
- By the rule of alligation, we have :
- C.P. of 1kg dal of 1st kind　　C.P. of 1kg dal of 2nd kind



x/y = 3/2= 3:2

Ans: A

Mixtures & Alligation

Q. A person blends two varieties of tea, one cost Rs. 160/kg and other cost Rs. 200/kg in the ratio 5 : 4. He sells the blended variety at Rs.192/kg. Find the profit %.

A. 6% B. 8% C. 7% D. 9%

Soln :

$\frac{x}{y} = \frac{d-m}{m-c}$

$\frac{5}{4} = \frac{200-m}{m-160}$

5m -800 = 800 -4m

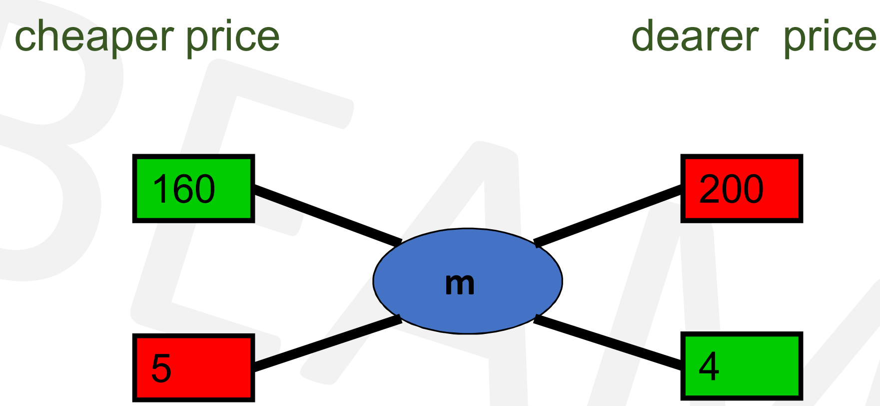
9m = 1600

m = $\frac{1600}{9}$

SP=Rs.192(given) , CP =mean price

Profit% = $\frac{SP-CP}{CP}$ x 100

$= \frac{192-\frac{1600}{9}}{\frac{1600}{9}} = \frac{1728-1600}{1600} = \frac{128}{16} = 8\%$ **Ans: B**



Mixtures & Alligation

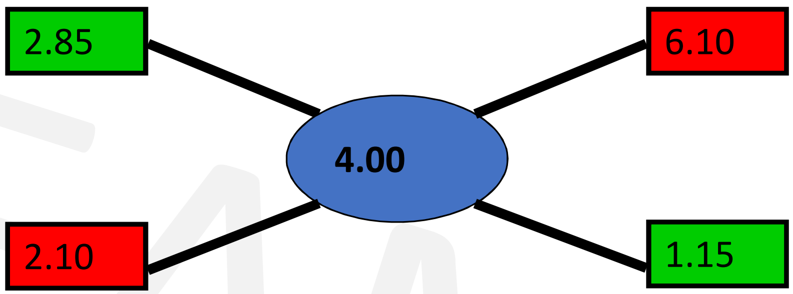
Q. What quantity of sugar costing Rs 6.10 per kg must be mixed with 126 kg of sugar priced at Rs 2.85 per kg so that 20% may be gained by selling mix at Rs 4.80/kg ?

A. 65 kg B. 69 kg C. 66 kg D. 64 kg

Soln :

$$SP = \frac{C.P. \times (100 + \%gain)}{100}$$

- CP (Mean) = 4.8 x 100/120 = 4.0
- Qty of Low : Qty of High = 210/115 = 42/23
- 126 / Qh = 42/23
- Ql = 126 x 23/42 = 69
- **Ans B**



Mixtures & Alligation

Q. Acid and water are mixed in a vessel A in the ratio of 5:2 and in the vessel B in the ratio of 8:5. In what ratio mixtures from two vessels should be mixed to get a new mixture in which the acid and water will be in the ratio of 9:4?

A. 7:2 B. 2:7 C. 7:4 D. 2:3

Soln: For these type of questions consider 1 ingredient out of the two ingredients and represent as fraction of one.

A
a:w
5:2

B
a:w
8:5

C
a:w
9:4

We consider acid here, so fraction of acid,

A
$\frac{5}{5+2} = \frac{5}{7}$

B
$\frac{8}{8+5} = \frac{8}{13}$

C
$\frac{9}{9+4} = \frac{9}{13}$

To make calculations easier, convert all denominator into common one

So, find LCM(7,13,13) = 91

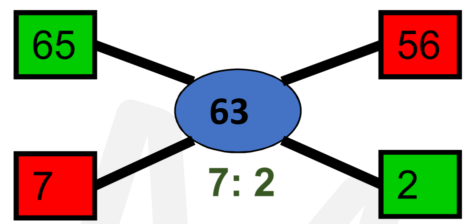
A
$\frac{5}{7}$ x $\frac{13}{13}$ = $\frac{65}{91}$

B
$\frac{8}{13}$ x $\frac{7}{7}$ = $\frac{56}{91}$

C
$\frac{9}{13}$ x $\frac{7}{7}$ = $\frac{63}{91}$

forget denominators,

By rule of Alligation,



Ans: A

Mixtures & Alligation

- **Final concentration = Initial $\left(1-\frac{R}{Initial}\right)^n$**
- where,
- Final concentration is the amount of concentration remaining after the process
- n is the number of times the process is done and
- R is the replaced quantity.
- Initial is the initial concentration

Mixtures & Alligation

Q. A container contains 60 litres of milk. From this container 6 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

A. 38.52 litres B. 43.74 litres C. 42.68 litres D. 39.85 litres

Ans: B

- The volume of milk remaining after the three processes is,
- $V=N(1-\frac{R}{N})^n$ where,

N is the original amount of milk,
n is the number of processes and
R is the replaced quantity.

$= 60(1-\frac{6}{60})^3$

$= 60(1-\frac{1}{10})^3$

$= 60(0.729)$

$= 43.74$ litres

Mixtures & Alligation(Assignment)

Q. A container contains 100 L of milk. From this container 10 L of milk was taken out and replaced by water. This process was further repeated three times. How much milk does the container have now?

A. 72.9 litres B. 65.61 litres C. 34.39 litres D. 81 litres

Ans: B

Final concentration = Initial concentration (1−Replaced/Initial)n

Mixtures & Alligation(Assignment)

Q. The ratio of milk to water in 80 litres of a mixture is 7 : 3. The water (in litres) to be added to it to make the ratio 2 : 1 is ?

A. 4 litres   B. 5 litres   C. 6 litres   D. 8 litres

Soln:

Mixture = 80 litres

Milk : Water

7 : 3 = 7+3 = 10(total parts of mixture)

Quantity of Milk = $\frac{7}{10}$ x 80 = 56 litres

Quantity of Water = $\frac{3}{10}$ x 80 = 24 litres

Let quantity of water added be 'x' litres

$\frac{56}{24+x} = \frac{2}{1}$ 56 = 48 + 2x

x = 4 litres of water is to be added.

OR

Let, Milk = 7x and Water = 3x

7x + 3x = 80 litres

10x = 80

x = 8litres

Milk = 7x = 7x8 = 56 litres

Water = 3x =3x 8 = 24 litres

$\frac{56}{24+x} = \frac{2}{1}$ 56 = 48 + 2x

x = 4 litres of water is to be added.

Ans : A

Mixtures & Alligation(Assignment)

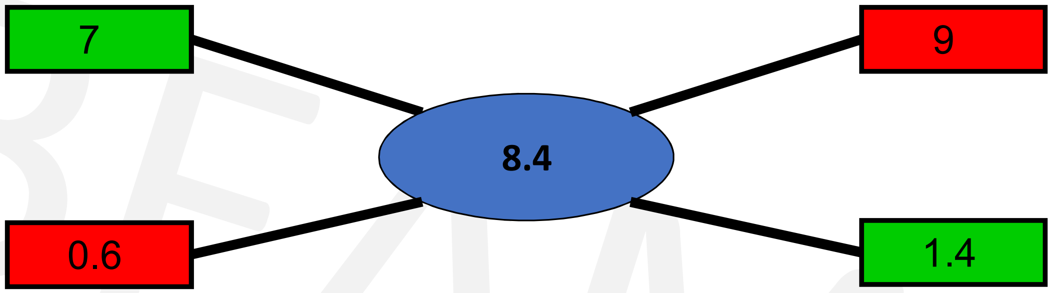
Q. How many kg of sugar costing Rs. 9 per kg must be mixed with 27kg of sugar costing Rs. 7 per kg, so that there maybe a gain of 10% by selling the mix at 9.24 per kg ?

A. 62kg B. 63kg C. 53kg D. 59kg

Soln:

$$SP = \frac{C.P. \times \mathbf{(100 + \%gain)}}{100}$$

CP (Mean) = 9.24 x 100/110 = 8.4



- Qty of Low : Qty of High = 0.6/1.4 = 6/14 = 3/7
- 27 / Q_H = 3/7
- Q_H = 27 x 7/3 = 63 kg

Ans: B

Mixtures & Alligation(Assignment)

Q. The cost of two varieties of tea is ₹300 and ₹375 respectively. If both the varieties of tea are mixed together in the ratio 3 : 2, then what should be the price of mixed variety of tea per kg?

A. Rs. 340 B. Rs.330 C. Rs. 350 D. Rs. 360

Ans: B

Mixtures & Alligation(Assignment)

Q. What quantity of sugar costing Rs 21.20 per kg must be mixed with 144 kg of sugar priced at Rs 26.20 per kg so that 10% may be gained by selling mix at Rs 25.30/kg ?

A. 256 kg   B. 265 kg   C. 244 kg   D. 144 kg

Ans: A

Mixtures & Alligation(Assignment)

Q. What ratio must a grocer mix tea at Rs. 60/ kg and Rs. 65/kg, so that by selling the mixture at Rs. 69.30/kg, he may gain 10%

A. 2:3 B. 4:3 C. 3:2 D. 2:1

Ans: A

Mixtures & Alligation(Assignment)

Q. Find the ratio in which the contains of 2 jars A & B containing spirit & water in the ratio 1:3 & 3:2 respectively must be mixed so that resulting mixture contains 45% spirit?

A. 2:3 B. 3:5 C. 3:2 D. 3:4

Ans D

Mixtures & Alligation(Assignment)

Q. Two solutions have milk : water ratio of 2:3 and 4:5. In what ratio must they be mixed such that the resultant solution has milk : water ratio of 3:4?

A. 8:3 B. 3:8 C. 5:9 D. 9:5

Ans : C

Mixtures & Alligation(Assignment)

Q. In what ratio rice at Rs. 9.30/kg be mixed with rice at Rs. 10.80/kg. So that the mixture be worth Rs. 10/kg.

A. 6:5 B. 8:7 C. 3:7 D. 6:1

Ans : B

Mixtures & Alligation(Assignment)

Q. The ratio, in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea when sold for Rs. 194.40 per kg, gives a profit of 20%.

A. 2 : 5 B. 3 : 5 C. 5 : 3 D. 5 : 2

Ans : A

Mixtures & Alligation(Assignment)

Q. In what ratio must a mixture of 30% alcohol strength be mixed with that of 50% alcohol strength so as to get a mixture of 45% alcohol strength?

A. 1 : 2 B. 1 : 3 C. 2 : 1 D. 3 : 1

Ans : B

Mixtures & Alligation(Assignment)

Q. A mixture of 70 litres of alcohol and water contains 10% of water. How much water must be added to the above mixture to make the water 12.5% of the resulting mixture?

A. 1 litre B. 1.5 litres C. 2 litres D. 2.5 litres

Ans: C

- Water=10% of 70 lit=7 lit,
- alcohol=90% of 70 lit=63 lit.
- Let, x lit water must be added.
  $\frac{(7+x)}{63} = \frac{12.5\%}{87.5\%}$
- 7 +x = 787.5/87.5
  7 + x = 9
- x=2 litres

Mixtures & Alligation(Assignment)

Q. In what ratio should two qualities of coffee powder having the rates of ₹47 per kg and ₹32 per kg be mixed in order to get a mixture that would have a rate of ₹37 per kg?

A. 1 : 2 B. 4 : 1 C. 1 : 3 D. 3 : 1 E. 1 : 4

Ans: A

Mixtures & Alligation(Assignment)

Q. How many kilograms of tea worth Rs. 3. 60 per kg. must be mixed with 8 kg. of tea worth Rs. 4.20 per kg. so that by selling the mixture at Rs. 4.40 per kg. There may be a of 10%.

A) 4 kg B) 3 kg. C) 6 kg. D) 8 kg.

Ans: A

Mixtures & Alligation(Assignment)

Q. The ratio of milk to water in 20 litres of a mixture is 3 :1. The Milk (in litres) to be added to the mixture so as to have milk and water in the ratio 4 : 1 is ?

A. 7 litres B. 4 litres C. 5 litres D. 6 litres

Ans: C

Mixtures & Alligation(Assignment)

Q. In what ratio must water be mixed with milk costing Rs. 12 per litre to obtain a mixture worth of Rs. 8 per litre?

A. 1 : 2 B. 2 : 1 C. 2 : 3 D. 3 : 2

Ans: A

By the rule of alligation :

$$\begin{array}{ccc} 0 & & 12 \\ & 8 & \\ 4 & : & 8 \end{array}$$

Ratio of water to milk

= 4 : 8

= 1 : 2

Mixtures & Alligation(Assignment)

Q. Two jars A and B contain milk and water in the ratio 7:5 and 17:7 respectively. In what ratio mixtures from two vessels should be mixed to get a new mixture containing milk and water in the ratio 5:3?

A. 2:1 B. 1:2 C. 2:3 D. 3:4

Soln:

For these type of questions consider 1 ingredient out of the two ingredients and represent as fraction of one.

A B

m:w m:w

7:5 17:7

C

m:w

5:3

We consider milk here, so fraction of milk,

A

$\frac{7}{7+5} = \frac{7}{12}$

B

$\frac{17}{17+7} = \frac{17}{24}$

C

$\frac{5}{5+3} = \frac{5}{8}$

To make calculations easier, convert all denominator into common one

So, find LCM(12,24,8) = 24

A

$\frac{7}{12} \times \frac{2}{2} = \frac{14}{24}$

B

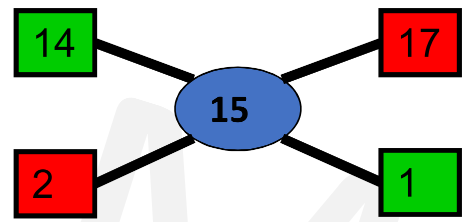
$\frac{17}{24}$

C

$\frac{5}{8} \times \frac{3}{3} = \frac{15}{24}$

forget denominators,

By rule of Alligation,



2:1

Ans: A

Mixtures & Alligation(Assignment)

Q. Two vessels A and B contain spirit and water mixed in the ratio 5:2 and 7:6 respectively. Find the ratio in which these mixtures are mixed to obtain a new mixture in vessel C containing spirit and water in the ratio 8:5?

A. 4:3

B. 3:4

C. 5:6

D. 7:9

Ans: D

Mixtures & Alligation(Assignment)

Q. A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

A. 26.34 litres B. 27.36 litres C. 28 litres D. 29.16 litres

Ans: D

- The volume of milk remaining after the three processes is,
- $V=N(1-\frac{R}{N})^n$ where, N is the original amount of milk, n is the number of processes and R is the replaced quantity.

$= 40(1-\frac{4}{40})^3$

$= 40(1-\frac{1}{10})^3$

$= 40(0.729)$

$=29.16$

Profit & Loss

- **Basics**

Profit (Gain) = (S.P – C.P)

Loss =(C.P – S.P)

% gain = (Gain / C.P) x 100

% loss = (Loss / C.P) x 100

- **Multipliers to find S.P**

In Case of Profit : S.P. = C.P. x **(100 +%gain)/100**

In Case of Loss : S.P. = C.P. x **(100 - %loss)/100**

i.e For sale at 25% profit S.P. = 125 % of C.P.

For sale at 25% loss S.P. = 75% of C.P.

Profit & Loss

Q. A man bought certain no of oranges at the rate of 5 for Rs 4 and sold them at the rate of 4 for Rs 5. Find his overall profit/loss percentage?

A. 25.5% Pr   B. 36.5% Pr   C. 56.2% Pr   D. 64.5% Pr

Soln

Cost Price		Selling Price	
Oranges→	Rs	Oranges→	Rs
5 →	4	4 →	5
20 →	16	20 →	25

SP>CP, so profit

P% = (SP –CP)/CP x 100

= (25-16)/16 x 100

= 225/4 = 56.20%

Ans: C

Cost Price		Selling Price	
Oranges→	Rs	Oranges→	Rs
5 →	4	4 →	5
1 →	$\frac{4}{5}$	1 →	$\frac{5}{4}$

SP>CP, so profit

P% = (SP –CP)/CP x 100

$$= \frac{\left(\frac{5}{4} - \frac{4}{5}\right)}{\frac{4}{5}} \times 100 = \frac{\left(\frac{9}{20}\right)}{\frac{4}{5}} \times 100$$

= 225/4 = 56.20%

Profit & Loss

Q. A man bought banana at the rate of 8 for Rs 34 and sold them at the rate of 12 for Rs 57 How many banana should be sold to earn a net profit of Rs. 45?

A. 90 B. 100 C. 135 D. 150

Soln:-

- Cost Price — Selling Price
- banana → Rs — banana → Rs
  - 8 → 34 — 12 → 57
  - 1 → $\frac{34}{8} = \frac{17}{4}$ — 1 → $\frac{57}{12} = \frac{19}{4}$

- SP>CP, so profit
- Profit = (SP –CP)
- $= \frac{19}{4} - \frac{17}{4} = \frac{1}{2}$

No. of banana to make a profit of Rs.45 $= \frac{\text{Profit total}}{\text{Pofit one}} = \frac{45}{1/2}$ = 90 banana

Ans: A

Profit & Loss

Q. When a article is sold for Rs.116,the profit percent is thrice as much as when it is sold for Rs. 92. The cost price of article is –

A. Rs. 68 B. Rs. 72 C. Rs. 78 D. Rs. 80

Soln:

- %P when SP is 116 = 3 x %P when SP is 92
- $\frac{116-CP}{CP}$ x 100 = 3 x $\frac{92-CP}{CP}$ x 100
- 116 – CP = 276 – 3CP
- 2CP =160
- CP =Rs. 80
- **Ans: D**

Profit & Loss

Q. By selling 90 ball pens for Rs.160, a person loses 20%. How many ball pens should be sold for Rs.96, so as to have a profit of 20% ?

A. 20 B. 30 C. 36 D. 26

Ans: C

Soln:

SP =Rs. 160 ,loss =20%

CP = $\frac{100}{100-2}$ x 160 = Rs.200

SP =Rs. 96 ,profit =20%

CP = $\frac{100}{100+20}$ x 96 = Rs.80

Rs.200 CP of = 90 ball pens

Rs. 80 CP of = ? $\frac{90 \text{ x } 80}{200}$ = 36 ball pens

Profit & Loss

Q. If selling price is doubled, the profit triples. Find the profit %.

A. $66\frac{2}{3}\%$ B. 100% C. $105\frac{1}{3}\%$ D. 120%

Soln:

Let, CP = C , SP=S

As they ask profit % , we know profit = SP – CP

As per given,

3(S-C) = 2S-C

3S -3C = 2S-C

S = 2C

But, Profit = S –C = 2C – C = C

Profit % = $\frac{\text{profit}}{\text{CP}}$ x 100 = $\frac{C}{C}$ x 100 = 100%

Ans : B

Combination

Q. A merchant has 1000 kg of sugar, part of which he sells at 8% profit and rest at 18% profit. he gains 14% on the whole. What is the quantity sold at 18% profit ?

A. 300 kg B. 700 kg C. 600 kg D. 400 kg

Ans : C

Combination

Q. A man has 80 pens. He sells some of these at 15% profit and the rest at 10%loss. Overall he gets a profit of 10%. Find how many pens were sold at 15% profit ?

A. 16 B. 64 C. 40 D. 72

Ans: B

Mixtures & Alligation

Q. A person blends two varieties of tea, one cost Rs. 160/kg and other cost Rs. 200/kg in the ratio 5 : 4. He sells the blended variety at Rs.192/kg. Find the profit %.

A. 6% B. 8% C. 7% D. 9%

Soln :

$\frac{x}{y} = \frac{d-m}{m-c}$

$\frac{5}{4} = \frac{200-m}{m-160}$

5m -800 = 800 -4m

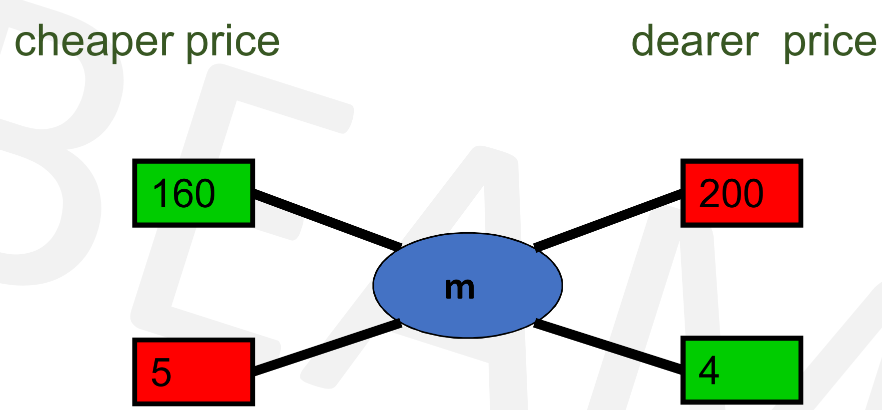
9m = 1600

m = $\frac{1600}{9}$

SP=Rs.192(given) , CP =mean price

Profit% = $\frac{SP-CP}{CP}$ x 100

$= \frac{192-\frac{1600}{9}}{\frac{1600}{9}} = \frac{1728-1600}{1600} = \frac{128}{16} = 8\%$ **Ans: B**



Profit & Loss

- **Marked price (MP)**
- A merchant, while selling goods, adds a certain percentage to the cost price (CP). This addition is called **percentage mark-up**, and the resulting price is called the **marked price (MP)**.
- Formulas:-
- Marked Price = CP + Mark Up
- Marked Price = CP + (%Mark Up on CP)
- Selling Price (SP) = Marked Price − Discount
- SP = MP × (100−Discount%) / 100 (if discount % is given)
- Discount % = (Discount / Marked Price) ×100
- CP = SP × 100 / (100+%Gain)

Profit & Loss

Q: The marked price of a book is ₹2,400. A bookseller gives a discount of 15% on it. What will be the cost price (₹) if he still earns a 20% profit?

A. Rs 1800 B. Rs 1700 C. Rs 1500 D. Rs 1600

Solution:

SP = MP × (100−Discount%) / 100

SP = 2400 × (100−15) / 100

SP = 2040

CP = SP × 100 / (100+%Gain)

CP = 2040 × 100 / (100+ 20)

CP = 1700

Ans: B

Profit & Loss(Assignment)

Q: A shopkeeper allows a discount of 10% on the marked price but still gains 20%. What would be the gain percentage if the article is sold at the marked price?

A. 25% B. 33.33% C. 35% D.50%

Ans: B

Profit & Loss(Assignment)

If gain is half of SP, the gain percentage is ____?

A. 50% B. 33.33% C. 25% D. 100%

Soln:

we know profit = SP – CP

As per given,

1/2SP = SP-CP

CP = SP – 1/2SP

SP = 2CP

But, Profit = SP –CP = 2CP– CP= CP

Profit % = $\frac{\text{profit}}{\text{CP}}$ x 100 = $\frac{\text{CP}}{\text{CP}}$ x 100 = 100%

Ans : D

Profit & Loss(Assignment)

Q. If the cost price of 6 pencils is equal to the selling price of 5 pencils, then the gain per cent is

A. 10% B. 20% C. 15% D. 25%

Soln:

Let the cost price of one pencil be Rs.1.
CP of 5 pencils =Rs. 5

CP of 6 pencils =Rs. 6

as, SP of 5 pencils = CP of 6 pencils

SP of 5 pencils = Rs.6
if, SP >CP so it's a profit
profit = SP − CP
= 6 − 5
= 1
Profit % = profit/cp x 100

= 1/5 x100

= 20%

SP=CP + gain
6 times CP is equal to 5 times SP

6CP=5SP
6CP=5(CP + gain)
6CP=5CP+5gain
CP=5 gain

Gain % = gain/CP x100
= 1/5 x 100
= 20%

Ans: B

Profit & Loss(Assignment)

Q. Ram bought 1600 eggs at the rate of Rs. 3.75 per dozen. He sold 900 of them at 2 eggs for Rs. 1 and the remaining at 5 eggs for Rs. 2. Find his loss or gain per cent.

A. 4% B. 45% C. 42% D. 46%

Ans: D

Price of 12 eggs = Rs. 3.75
Price of 1 egg = 3.75/12 = Rs.0.3125
Price of 1600 eggs = Rs. (0.3125 × 1600) = Rs.500
CP of eggs = Rs. 500
Selling price of 2 eggs = Rs. 1
Selling price of 900 eggs =
Selling price of 5 eggs = Rs. 2
Selling price of remaining (1600 - 900 = 700) eggs
Total SP of eggs = Rs. 450 + 280 = Rs.730.
SP > CP → profit/gain
Gain = SP – CP = Rs. 730 - 500 = Rs. 230.
%Gain = 230/500 x100 = 46%

Profit & Loss(Assignment)

Q. A shopkeeper purchases 11 sword for Rs.10 and sells them at the rate of 10 sword for Rs. 11. He earns a profit % of?

A. 11% B. 15% C. 20% D. 21%

Ans: D

Profit & Loss(Assignment)

Q. A bookseller sells 84 books at the cost of 72 books. Find his profit or loss%

A. 14.28% B. 28.24% C. 20.4% D. 12.86%

Ans : A

Profit & Loss(Assignment)

Q. By selling 100 pencils, a shopkeeper gains the selling price of 20 pencils. His gain per cent is

A) 25 B) 20 C) 15 D) 12

Ans: A

SP –CP =gain here gain = SP of 20 pencils

S.P. of 100 pencils − C.P. of 100 pencils = S.P. of 20 pencils
S.P. of 80 pencils = C.P. of 100 pencils

Let C.P. of 1 pencil = Rs. 1
S.P. of 80 pencils = Rs. 100

C.P. of 80 pencils = Rs. 80

Profit % = $\frac{100-8}{80}$ ×100 = 25%

Profit & Loss(Assignment)

Q. A man bought a horse & carriage together for Rs 15600 & sold them together, the horse at 36% profit & the carriage at 15% loss. If selling price of both is equal. Find the cost of the carriage?

A.Rs.6000 B. Rs.7600 C. Rs.3600 D. Rs.9600

- **Soln**
- Let CP of horse be H & Carriage be C → H+C= 15600
- SP of both is equal
- So, comparing the CPs
- 136H/100 = 85C/100
- H = 5C/8
- 5C/8 + C = 15600
- 13C/8 = 15600
- C = 1200 x 8
- C = 9600

Ans: D

Profit & Loss(Assignment)

Q. A vendor bought 6 oranges for Re 10 and sold them at 4 for Re 6. Find his loss or gain percent.

A. 8% gain B. 10% gain C. 8% loss D. 10% loss

Ans: D

Profit & Loss(Assignment)

Q. A shopkeeper sells his goods at 10% loss but uses a weight of 750gms instead of 1kg. Find profit %

A. 20% Pr B. 14.28% Pr C. 30% Pr D. 25% Ls

Ans: A

Profit & Loss(Assignment)

Q. A fruit seller buys oranges at 4 for Rs. 3 and sells them at 3 for Rs. 4. Find its profit percent.

A. 43.75% Pr B. 77.7% Pr C. 75% Pr D. 65.7% Ls

Ans: B

Profit & Loss(Assignment)

Q. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?

A. Rs. 1090 B. Rs. 1160 C. Rs. 1190 D. Rs. 1202

Ans: C

Profit & Loss(Assignment)

Q. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is:

A. 14 2/7% gain B. 15% gain C. 14 2/7% loss D. 15 % loss

Ans: A

Profit & Loss(Assignment)

Q. A shopkeeper sells his goods at 20% profit and to make an extra profit he gives only 800 gm per kg. Find his profit %

A. 25% Pr B. 33.33% Pr C. 50% Pr D. 25% Ls

Soln

CP	SP	Profit
100	120	20
80	120	40

% Profit = 40/80 x 100

= 1/2 x 100

= 50%

Ans: C

Probability

- How likely an event is supposed to happen.
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}}$
- AND → multiply(x) e.g:- 1green and 1 blue ball in a box
- OR → Add (+) e.g:- 1 red or 1 blue ball in a box
- 1 bag has 3 balls, what is the probability of you picking up 2 balls?
    - $3C_2 = \frac{3\text{x } 2}{1 \text{ x } 2} = 3$

Total no. of balls
the bag contains

Out of which how many balls
We need to choose
(tells number of times 3 has to be reduces)

Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}}$

Points to Remember

- The **probability** of an event will not be less **than** 0.
- This is because 0 is impossible (sure that something will not happen).
- The **probability** of an event will not be **more than 1**. This is because **1** is certain that something will happen.
- The probability of an event is **a number** describing the chance that the event will happen.
- An event that is certain to happen has a probability of 1.
- An event that cannot possibly happen has a probability of 0.
- If there is a chance that an event will happen, then its probability is between 0 & 1.

Probability

- **Atleast** – min to max
- Eg:- 2 bags out of 3

min max

So various probabilities to be done is 2 and 3

- **Atmost** - max to min
- Eg:- A bag has 3 balls out of which probability to pick up 2 balls

atmost 2 → max 2 , 1 , 0 (min)

Probability

Q. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

A. 10/21 B. 11/21 C. 2/7 D. 5/7

- **Soln-**
- Total balls = 2+3+2 =7 balls in the bag
- None = blue (neglect whichever color is written after none)
- Draw =2 balls
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{2\text{R or } (1\text{R and } 1\text{ G}) \text{ or } 2\text{G}}{7_{C_2}} = \frac{2C_2 + (2C_1 \times 31) + 3C_2}{7C_2} = \frac{10}{21}$

Ans : A

Probability

Q. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?

A. 1/3 B. ¾ C. 7/19 D. 8/21 E. 9/21

Soln:

- Total balls = 8+7+6 = 21 balls in the box
- Neither red nor green means only blue
- Draw =1 ball
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{1 blue\ out\ of\ total\ 7}{21\ \ _1} = \frac{7C_1}{21C_1} = \frac{7}{21} = \frac{1}{3}$

Ans: A

Probability

Q. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:

A. 21/46 B. 25/117 C. 1/50 D. 3/25

Soln:

- Total students = 15 + 10 =25 students in a class
- Draw =3 students

Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{10C_1 \text{ x } 15\ {}_2}{25C_3} = \frac{21}{46}$

Ans : A

Probability

Q. What is the probability of getting a sum 5 from two throws of a dice?

A. 1/9 B. 1/8 C. 1/7 D. 1/6

Soln-

Dice =6 faces = 6 possibilities

So in two throws of dice, total possibilities = **6 x 6= 36**

Sum =5,so favourable outcomes are - { (1,4), (4,1) , (2,3) , (3,2) }

$$\text{Probability} = \frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{4}{36} = \frac{1}{9}$$

Ans : A

Probability

Q. Three unbiased coins are tossed. What is the probability of getting atmost two heads?

A. ¾ B. ¼ C. 3/8 D. 7/8

- **Soln-**
- Total possibilities = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}
- Event of getting utmost 2 heads = max 2H or 1H or 0H
- Possibility of getting 2 H = {TTT, TTH,THT, HTT, THH, HTH, HHT}
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{7}{8}$

Ans: D

Probability

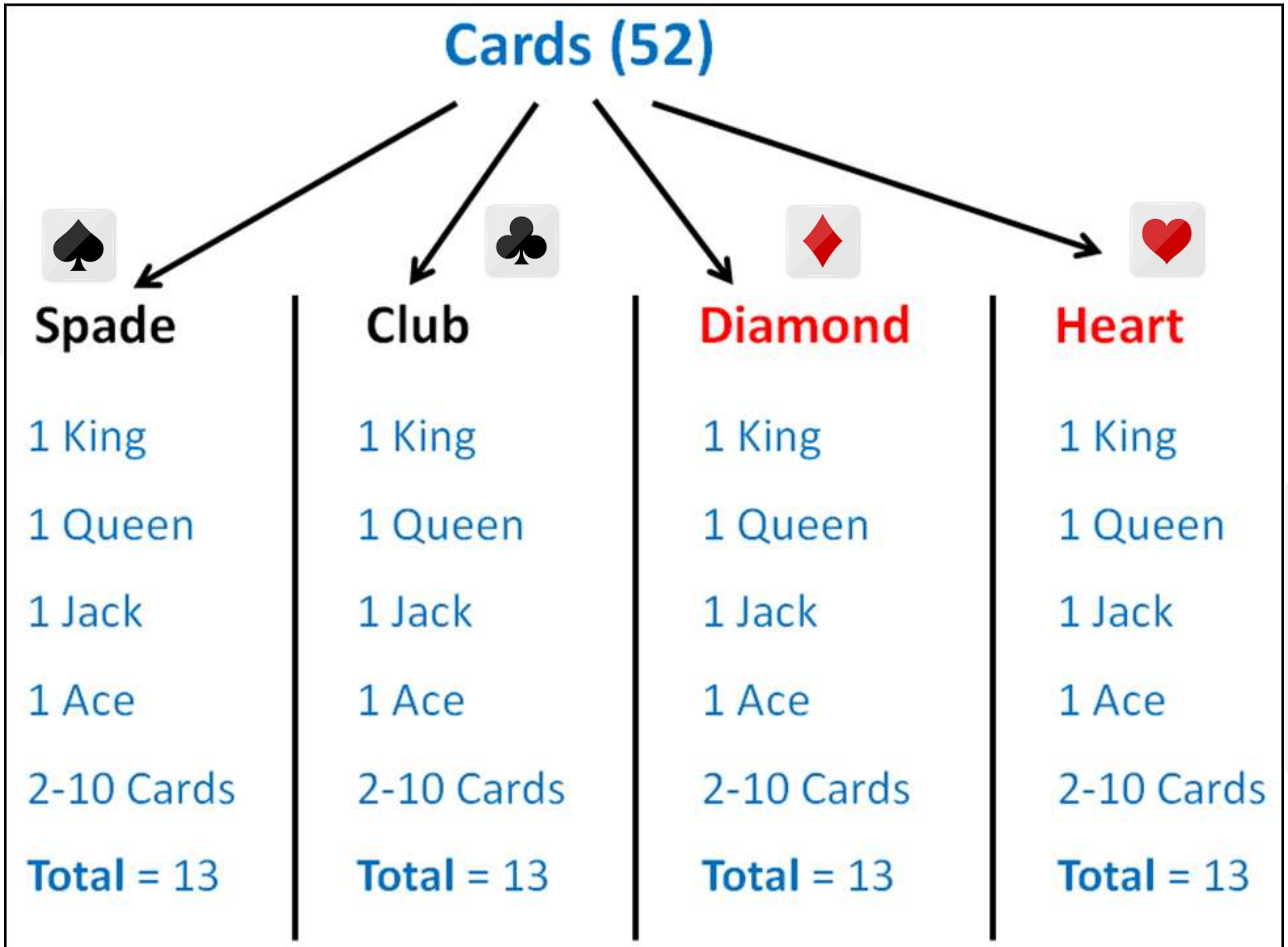
- A Standard deck of playing cards consist of 52 cards, among them there are 4 subgroups/suits –
- The four suits with there names , symbols and color –

1. The suit of Hearts  → 13 cards
2. The suit of Diamonds  → 13 cards
3. The suit of Clubs  → 13 cards
4. The suit of Spades  → 13 cards

26 red cards

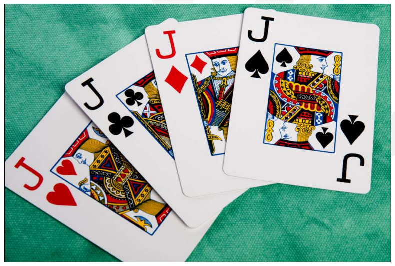
26 black cards

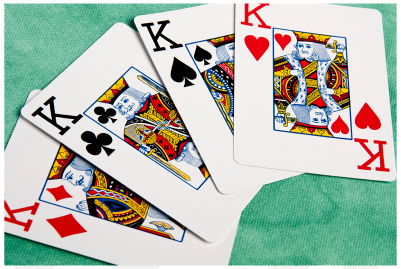
Probability



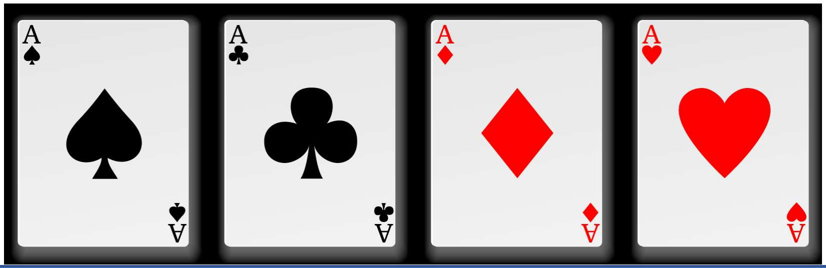
Probability

- King, Queen and Jack (or Knaves) are **face cards**. So, there are **12 face cards** in the deck of 52 playing cards.
- **Jokers** are not normally considered to be **face cards**





- **Aces**
- There are 4 Aces in every deck, 1 of every suit.



Probability

Q. From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?

A. 1/15 B. 25/57 C. 35/256 D. 1/221

- **Soln-**
- **Total cards in a pack =52**
- Total kings in a pack = 4
- Drawn =2
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{4_{C_2}}{52_{C_2}} = \frac{1}{221}$

Ans : D

Probability

Q. Out of 450 students of a school 325 play football,175 play cricket and 50 neither play football nor cricket. How many students play both football and cricket?

A. 50 B. 75 C. 100 D. 225

Soln:

- $A \cup B$ is used for the set that contains those elements that are either in A or in B, or in both.
- $A \cap B$ is used for the largest set which contains all the elements that are common to both the sets.
- **$n(A \cap B) = n(A) + n(B) - n(A \cup B)$ → addition theorem of probability.**
- Total students who play either Cricket or football or both cricket and football
- $= n(A \cup B) = 450 - 50 = 400$
- students who play cricket, $n(A) = 325$ and students who play football, $n(B) = 175$
- Students who play both football and Cricket $= n(A \cap B) = 325 + 175 - 400 = 100$
- **Ans: C**

Probability(Assignment)

Q. Two dice are rolled. Find the probability of getting a sum of 8 or 11 on both the dices.

A. 5/36 B. 9/36 C. 7/36 D. 11/36

Ans: C

- Favorable outcomes for sum of 8 or 11 on both the dices are-
- (2,6),(3,5),(4,4),(5,3),(6,2),(5,6),(6,5)
- Number of favorable outcomes = 7
- Probability = $\frac{7}{36}$

Probability(Assignment)

A man tossed two dice. What is the probability that the total score is a prime number?

A. 5/12 B. 5/14 C. 5/20 D. 5/24

- **Soln-**
- **Dice =6 faces = 6 possibilities**
- 2 Dice = 6 x 6 = 36 possibilities
- Sum = prime number
- So favourable outcomes are - { (1,1), (1,2) , (1,4), (1,6) , (2,1) , (2,3) , (2,5) , (3,2) , (3,4) , (4,1) , (4,3) , (5,2) , (5,6) , (6,5) , (6,1) }
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{15}{36} = \frac{5}{12}$

Ans : A

Probability(Assignment)

Q. A brother and sister appear for an interview against two vacant posts in an office. The probability of the brother's selection is 1/5 and that of the sister's selection is 1/3. What is the probability that one of them is selected?

A. 1/5 B. 2/5 C. 1/3 D) 2/3

Soln: -

(brother is selected and sister is not selected) OR (brother is not selected and sister is selected)

$$\text{Probability} = \frac{1}{5} \times \frac{2}{3} + \frac{4}{5} \times \frac{1}{3}$$

$$= \frac{6}{15}$$

$$= \frac{2}{5}$$

sister not selected = 1 – prob. of sister selected

$$= 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

brother not selected = 1 – prob. of brother selected

$$= 1 - \frac{1}{5}$$

$$= \frac{4}{5}$$

Ans: B

Probability(Assignment)

Q. Probability of occurrence of event A is 0.5 and that of event B is 0.2. the probability of occurrence of both A and B is 0.1. what is the probability that none of A and B occur?

A. 0.4 B. 0.5 C. 0.2 D. 0.1

Soln:

probability of sure event = 1

- Given P(A) = 0.5 and P(B) = 0.2
- P(A or B) = $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.5 + 0.2 - 0.1 = 0.6$$

- And P(neither A nor B) = $P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.6 = 0.4$.

Ans: A

- Note: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- This is also known as the addition theorem of probability.

Probability(Assignment)

Q. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is?

A. 1/22 B. 3/22 C. 2/91 D. 2/77

Ans : C

Probability(Assignment)

Q. What is the probability of getting a sum 9 from two throws of a dice?

A. 1/6 B. 1/8 C. 1/9 D. 1/12

Ans : C

Probability(Assignment)

Q. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?

A. ¾ B. 4/7 C. 1/8 D. 3/7

Ans : B

Probability(Assignment)

Q. A bag contains 6 blue balls, 3 white balls and 4 green balls. If two balls are drawn at random what is the possibility that they are not of the same color?

A. 6/13 B. 7/13 C. 9/13 D. 10/13

- **Ans: C**

Probability(Assignment)

Q. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card (Jack, Queen and King only)?

A. 1/13 B. 1/4 C. 3/13 D. 9/52

Ans: C

Probability(Assignment)

Q. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is not a face card (Jack, Queen and King only)?

A. 5/13 B. 10/13 C. 1/13 D. 1/26

Ans: B

Probability(Assignment)

Q. A basket contains 6 apples ,4 pears and 3 oranges. If two fruits are picked up at random, what is the probability that both are pears?

A. 4/13 B. 1/13 C. 2/13 D. 3/26

Ans: B

Partnership

Q.A started business with Rs. 45,000 and B joined afterwards with 30,000. If the profit at the end of a year was divided in the ratio 2 : 1 respectively, then B would have joined A for business after.

A. 1 month B. 2 months C. 3 months D. 4 months

Soln:

- Capital of A = Rs. 45,000 Capital of B = Rs. 30,000
- Ratio of P1:P2=2:1
- using formula,
- $\frac{C1T1}{C2T2} = \frac{P1}{P2}$
- In this type , the time period is 12 months i.e. one year
- $\frac{45000 \times 12}{30000 \times T2} = \frac{2}{1}$
- T2=9
- B would join business after (12 - 9) = 3 months
- **Ans: C**

THANK YOU!